

Task 21 – Basic SQL SELECT

Data Analytics Track | Level 1 – Day 21

Objective

Build SQL fundamentals by writing simple SELECT, WHERE, and ORDER BY queries and understanding how SQL retrieves and sorts data.

Tools

MySQL / PostgreSQL concepts were followed. The completed task also includes a local SQLite database for easy testing.

Dataset

A Northwind-style Products table was used with fields such as ProductID, ProductName, Category, UnitPrice, and UnitsInStock.

Queries Completed

No.	SQL Query
1	<code>SELECT * FROM Products;</code>
2	<code>SELECT ProductName, UnitPrice FROM Products;</code>
3	<code>SELECT * FROM Products WHERE Category = 'Beverages';</code>
4	<code>SELECT * FROM Products WHERE UnitPrice > 20;</code>
5	<code>SELECT * FROM Products WHERE UnitsInStock < 20;</code>
6	<code>SELECT ProductName, UnitPrice FROM Products ORDER BY UnitPrice ASC;</code>
7	<code>SELECT ProductName, UnitPrice FROM Products ORDER BY UnitPrice DESC;</code>
8	<code>SELECT * FROM Products WHERE Category = 'Condiments' ORDER BY ProductName ASC;</code>
9	<code>SELECT ProductName, UnitsInStock FROM Products WHERE UnitsInStock > 30 ORDER BY UnitsInStock DESC;</code>
10	<code>SELECT ProductName, UnitPrice, UnitsInStock FROM Products WHERE UnitPrice BETWEEN 10 AND 50 ORDER BY UnitPrice ASC;</code>

Key Concepts Demonstrated

- SELECT retrieves columns or rows from a table.
- WHERE filters records according to a condition.
- ORDER BY sorts the result set in ascending (ASC) or descending (DESC) order.
- BETWEEN filters values within an inclusive range.
- Multiple clauses can be combined to filter and sort results.

Sample Output / Validation

The queries were organized in a separate SQL script and validated against the included sample database. Output files are provided with the task ZIP so the results can be reviewed query-by-query.

Interview Questions

1. What does **SELECT** do?

SELECT is used to retrieve data from one or more database tables.

2. What does **ORDER BY** do?

ORDER BY sorts query results based on one or more columns. **ASC** is ascending and **DESC** is descending.

3. What is **WHERE** used for?

WHERE filters rows before the result is returned.

4. What is the difference between **SELECT *** and selecting columns?

SELECT * returns all columns, while specifying columns returns only the required fields.

Conclusion

Task 21 was completed to practice the core SQL retrieval workflow: selecting required data, filtering records with **WHERE**, and sorting results with **ORDER BY**. These are foundational skills for data analysis and database querying.